Evaluation Report of EliteFS-Day23-AI-KMIT

Average Score: 8/10 | Attempted: 3 of 3 questions

Question 1

THEORY UNDERSTAND Best Score: 7/10

Why should the weights in Xavier initialization be in the range –sqrt(6/(nin+nout)) to + sqrt(6/(nin+nout))?

Desired Answer

Xavier initialization chooses the range –sqrt(6/(nin+nout)) to + sqrt(6/(nin+nout)), so that the variance of the weights becomes 2/(nin+nout). This keeps the variance of activations and gradients approximately constant across layers. As a result, it prevents vanishing and exploding gradients during training. The specific range comes from matching the variance of a uniform distribution to this desired value.

Student's Answer

As this is a range of weights which are found to be optimal in initialization. Xavier Initialization is a smart way of random initialization. In Normal random initialization we have risk of weight being very high or very low leading to Exploding Gradient and Vanishing Gradient Problems respectively. But by doing Xavier Initialization, we can achieve a varience of 2/(nin+nout) which leads to all the weight being in a specific range and not be very random in nature.

Evaluations

Feedback 1

★★★☆☆ 6/10

The student is discussing the same Xavier initialization concepts as the Actual Answer (the specific range, achieving variance of 2/(nin+nout), and preventing vanishing/exploding gradients), so the response is on-topic. They correctly note that choosing weights in this range yields the desired variance and helps avoid gradient issues. However, they do not explain how the uniform distribution's variance is matched to 2/(nin+nout), nor do they explicitly connect this choice to keeping both activations and gradients stable across layers. Their answer remains at a somewhat superficial level, missing the derivation and fuller mechanism from the Actual Answer, so it demonstrates adequate but not deep understanding.

Feedback 2 ★★★½ 7/10

The student is addressing the same concepts as the Actual Answer: they explain that Xavier initialization picks a specific symmetric range to fix the variance of the weights at 2/(n_in + n_out), and they note that this stabilizes training by preventing vanishing and exploding gradients. This shows a clear grasp of the core idea. However, the answer is somewhat shallow: it does not detail how the ±√(6/(n_in + n_out)) bounds arise from matching a uniform distribution's variance to the target value, and the phrasing "not be very random in nature" is vague. Overall, the student demonstrates solid understanding of the main points but omits the derivation detail, fitting a score of 7.

Question 2

THEORY ANALYZE Best Score: 8/10

What is the difference between an eigenvector and an eigenvalue?

Desired Answer

An eigenvector is a special vector whose direction does not change when a matrix transformation is applied to it. An eigenvalue is the scalar that tells how much that eigenvector is stretched or compressed during the transformation. In the equation $Av=\lambda v$, v is the eigenvector and λ is the eigenvalue. In simple terms, the eigenvector gives the direction, and the eigenvalue gives the scaling factor in that direction. In simple terms: Eigenvector = special direction, Eigenvalue = amount of scaling in that direction

Student's Answer

Eigenvectors represent the direction in which a matrix is unchanged where as eigen values represent the magnitude of the eigen vectors. Eigen vectors are vectors and eigen value is a number. These are a backbone in ML for Principle Component Analysis (PCA)

Evaluations

Feedback 1 ★★★★ 8/10

The student is addressing the same concepts as the Actual Answer (eigenvectors vs. eigenvalues). They correctly identify that an eigenvector indicates a direction that remains invariant under the transformation, and that an eigenvalue is a scalar (number) associated with it. This matches the core idea that eigenvectors give direction and eigenvalues give the scale change. They miss explicitly stating that the vector is stretched or compressed by that factor (instead saying it represents the magnitude of the eigenvector) and do not reference the equation $Av = \lambda v$, but these are minor omissions. The extra mention of PCA is on-topic but not required. Overall, they demonstrate a solid understanding of the main concepts with only small gaps, meriting a score of 8.

Feedback 2 ★★★★☆ 8/10

The student is demonstrating understanding of the same concepts as the Actual Answer (the distinction between eigenvectors as directions preserved by a matrix transformation and eigenvalues as the associated scaling factors). They correctly state that eigenvectors are vectors and eigenvalues are scalars, and they capture the core idea that eigenvectors indicate direction and eigenvalues indicate magnitude changes. Their phrasing ('direction in which a matrix is unchanged') is slightly inaccurate (it's the vector's direction that remains unchanged), and they simplify eigenvalues as 'magnitude of the eigenvectors' rather than explicitly as 'stretching/compression factors.' However, these are minor imprecisions, and the student conveys most of the key ideas. Their extra note on PCA is supplementary and on-topic. This earns a high but not perfect score for good, mostly accurate understanding with small gaps in clarity.

Question 3

THEORY REMEMBER Best Score: 8/10

What is the role of gamma and beta in Batch Normalization?

Desired Answer

Gamma (γ) and Beta (β) are learnable parameters in Batch Normalization that allow the network to adjust the normalized output. After normalizing inputs to have zero mean and unit variance, γ scales the values and β shifts them. This ensures that the model can restore or modify the distribution if needed. Without γ and β, the normalization might limit the representational power of the layer.

Student's Answer

Gamma is the value used for scaling and beta is the value used for shifting in Batch Normalization. In batch normalization, we always reduce the values with mean 0 and varience 1 this leads for the signal to the next layers to be in that range only but sometimes we may need in different ranges.

Evaluations

Feedback 1

★★★☆☆ 6/10

The student correctly recalled that gamma scales and beta shifts the normalized output and that normalization produces zero mean and unit variance. They also hinted that different ranges may be needed (implying the use of γ and β to adjust the distribution). However, they failed to mention that γ and β are learnable parameters and did not explicitly state that they allow the network to restore or modify the distribution as needed (and that without them the layer's representational power would be limited). Therefore, key details from the Actual Answer are missing, yielding an Adequate Recall.

Feedback 2

★★★★☆ 8/10

The student correctly states that gamma performs scaling and beta performs shifting after normalizing inputs to zero mean and unit variance, and notes that this lets the network adjust the range as needed. They recall the core role of γ and β in adjusting the normalized output. They did not explicitly mention that γ and β are learnable parameters nor reference the representational-power rationale, but these are secondary details. Thus, most key facts from the Actual Answer are accurately recalled with minor omissions.